Sannidhi D Math

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ABOUT ME

Mechatronics graduate working on computer vision for robots and drones: perception and navigation for an agricultural robot, a detection-to-GPS system for drones, and ROS projects (URDF, Gazebo, Nav2). Co-author of two IEEE conference papers.

EDUCATION

Manipal Institute of Technology 2022–2026
B.Tech in Mechatronics Engineering (Minor: IIoT), GPA: 8.26/10.00

Base PU College, Bengaluru 2020–2022
Karnataka State Board 12th, Grade: 85%

SKILLS

Languages & Tools: Python, C++, C, SQL, Java; Linux (bash, ssh), Git, FastAPI, SQLite.

Computer Vision & ML: PyTorch, OpenCV, YOLOv5/v8, ResNet, ONNX Runtime, scikit-learn, transfer learning.

Robotics & ROS 2: ROS 2 Humble (nodes, services, launch files), URDF, Gazebo/RViz, ros2_control, Nav2, SLAM, PID control, visual odometry, path planning, sensor calibration.

RESEARCH & PUBLICATIONS

Manipal Institute of Technology, *Research Assistant* Feb 2024–Present

- *XGBoost-based Ensemble Regression for Building Energy Load Prediction*, CINS 2025, IEEE (RMSE 0.407 / 0.858).
- *Flood Susceptibility Modeling using Hybrid Machine Learning and GIS Techniques*, IEEE conference.

EXPERIENCE

Ather Energy March 2026–August 2026
Software Intern

- Sole owner of an LLM agent that finds root causes in ECU and CAN-bus logs, with a critic stage that checks conclusions against raw telemetry. Also built EV-charger certificate renewal and fault diagnostics.

Team RoboManipal Feb 2024–July 2025
Technical Member, Autonomous Agricultural Robot

- Worked on visual odometry, depth estimation and path planning for outdoor navigation, with sensor calibration and field trials. Trained a ResNet-18 plant classifier (92% accuracy).

IISc-TCS Innovation Lab June 2024–July 2024
Intern, Hyperspectral Imaging

- Interfaced a hyperspectral camera through Python APIs and built a K-Means/PCA pipeline on 10,000+ spectral signatures, improving classification accuracy by 15%.

PROJECTS

- **ROS Projects** — *ROS 2 Humble, Python, Gazebo, RViz, Nav2, OpenCV*
  Wrote ROS 2 nodes, a custom service and a go-to-goal controller. Modelled a mobile robot and a manipulator in URDF, simulated them in Gazebo with ros2_control, built an OpenCV line follower with PID, and ran SLAM and Nav2 on a TurtleBot3 in simulation.
- **Traffic Drone Geo-Referencing System** — *YOLOv8, ONNX Runtime, Leaflet.js* GitHub — Demo
  Trained YOLOv8 on VisDrone (9,287 images; mAP50 0.583) and converted detections to GPS coordinates from flight-log telemetry, camera geometry and heading. Clustered repeats within 5 m.